

CI/CD Tool Comparison Matrix

Criteria	GitHub Actions	Jenkins	GitLab CI
Setup	Zero setup – built into GitHub	Self-hosted, manual install required	Built into GitLab, minimal config needed
Cost	Free for public repos; 2000 min/month free for private	Free (open source), but server costs apply	Free tier available; paid for more minutes
Scalability	Auto-scales with GitHub-hosted runners	Manual scaling via agent nodes	Auto-scales with GitLab-hosted runners
Secrets	GitHub Secrets (encrypted, UI-managed)	Credentials plugin (Jenkins Credentials)	GitLab CI/CD Variables (masked/protected)
Community	Large, fast-growing; GitHub Marketplace	Massive, mature; 1800+ plugins	Strong, especially in enterprise DevOps
Config	YAML in .github/workflows/	Jenkinsfile (Groovy DSL)	YAML in .gitlab-ci.yml
Best For	GitHub-native projects, quick setup	Complex enterprise pipelines, full control	GitLab-hosted projects, end-to-end DevOps